

115 年度 社團法人中華產業機械設備協會 推廣費曼學習記錄

主講者 李益州

日期 115 年 3 月 10 日

1. 學習項目：複循環廢熱回收系統簡介

2. 學習動機：解析複式循環廢熱回收系統運作流程、運用熱力學流程圖解說明各部件運作運轉條件作為危險性設備檢查之參考

3. 績效與效益：

1. 於執行危險性設備檢查時對於系統運作流程溫度上之不同而設置之危險性設備功能及流程說明。
2. 瞭解對飼水系統所考量設計安全運轉條件及給水泵構造。
3. 提升危險性設備竣工檢查時之系統考量進而作為定期檢查之依據。

學習相片剪影





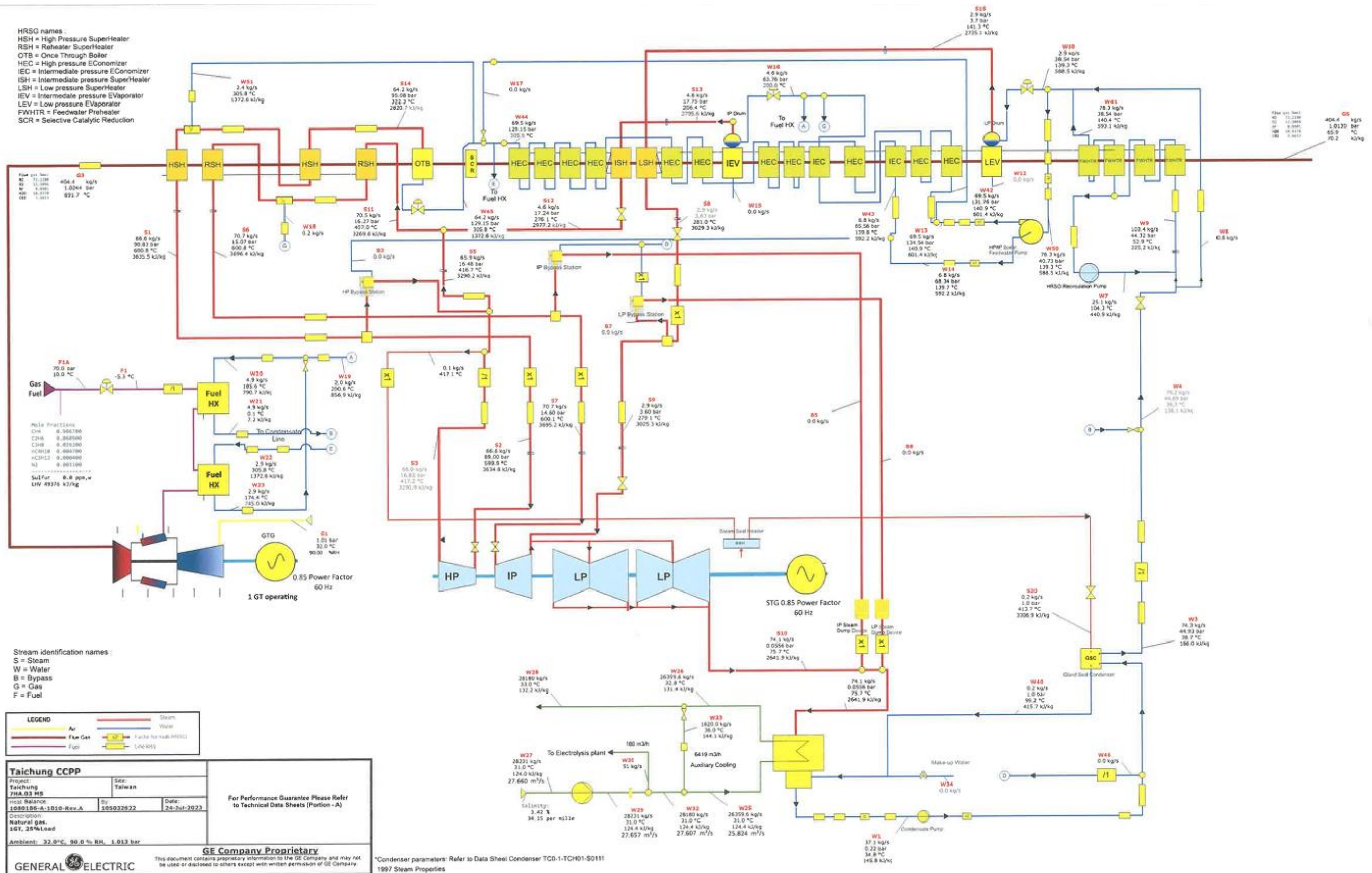
複循環廢熱回收系統

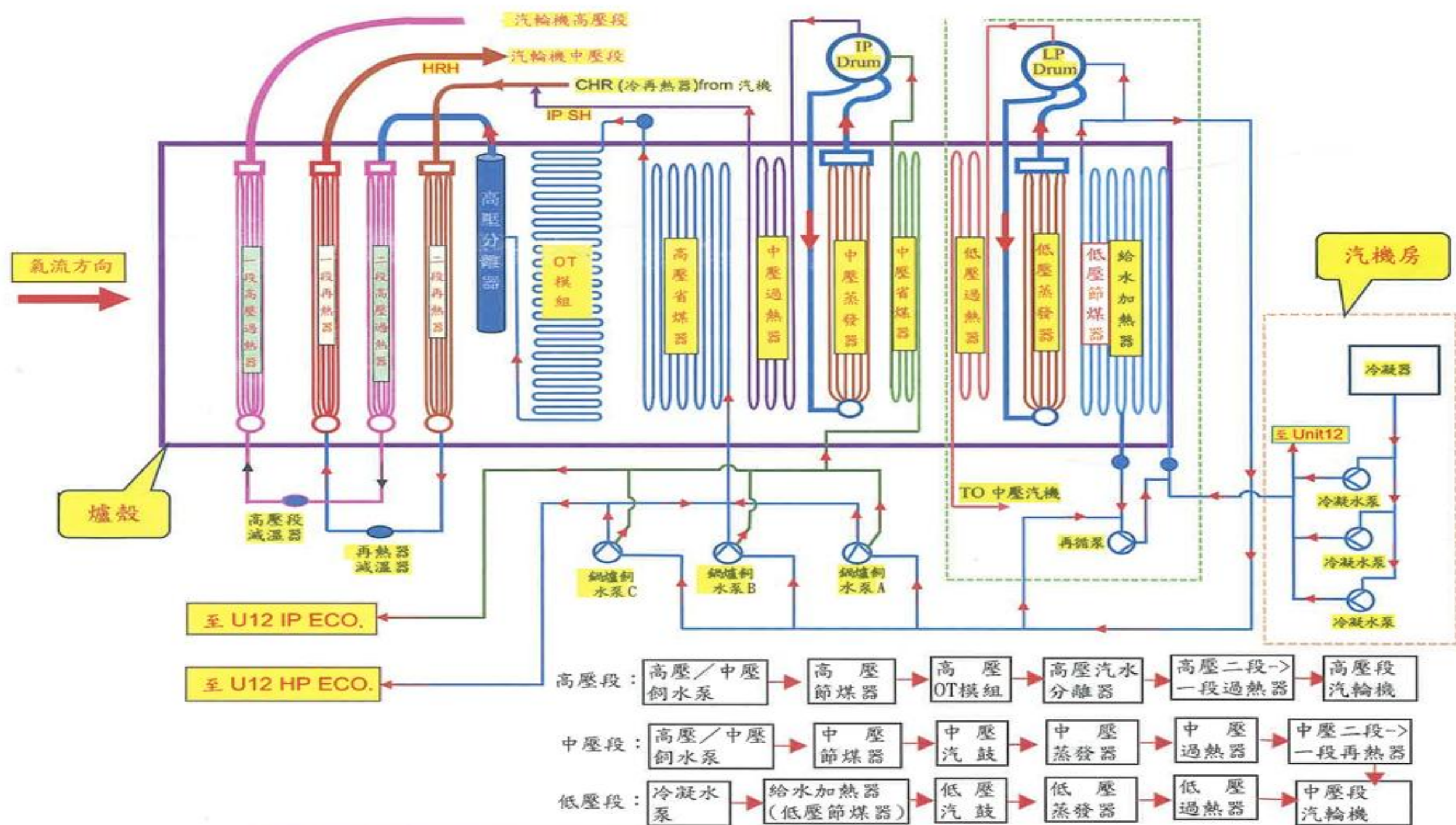
Combined Cycle Waste Heat Recovery System

運作原理解析與熱力學流程圖解

報告人：李益州 (Lee Yi-chou)

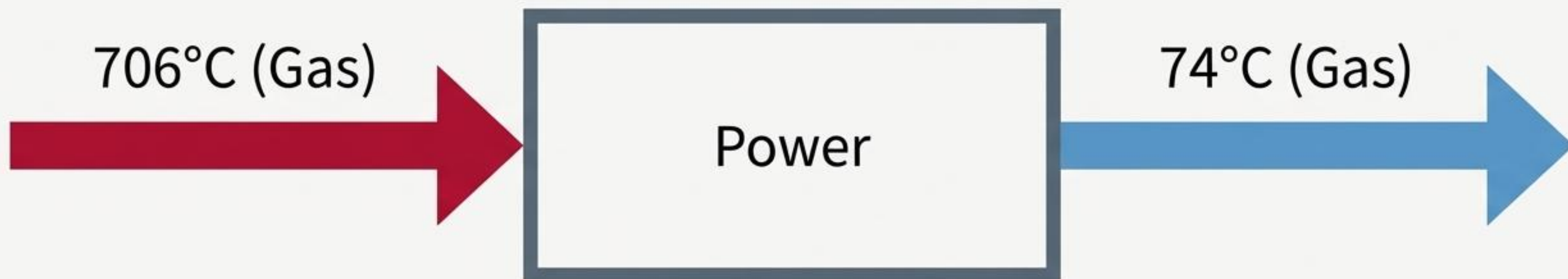
HRSG names:
 HSH = High Pressure SuperHeater
 RSH = Reheater SuperHeater
 OTB = Once Through Boiler
 HEC = High pressure EConomizer
 IEC = Intermediate pressure EConomizer
 ISH = Intermediate pressure SuperHeater
 LSH = Low pressure SuperHeater
 IEV = Intermediate pressure EVaporator
 LEV = Low pressure EVaporator
 FWHR = Feedwater Preheater
 SCR = Selective Catalytic Reduction





台中發電廠HRSG 複循環機組流程圖

系統核心概念：將極端廢熱轉化為實體動能



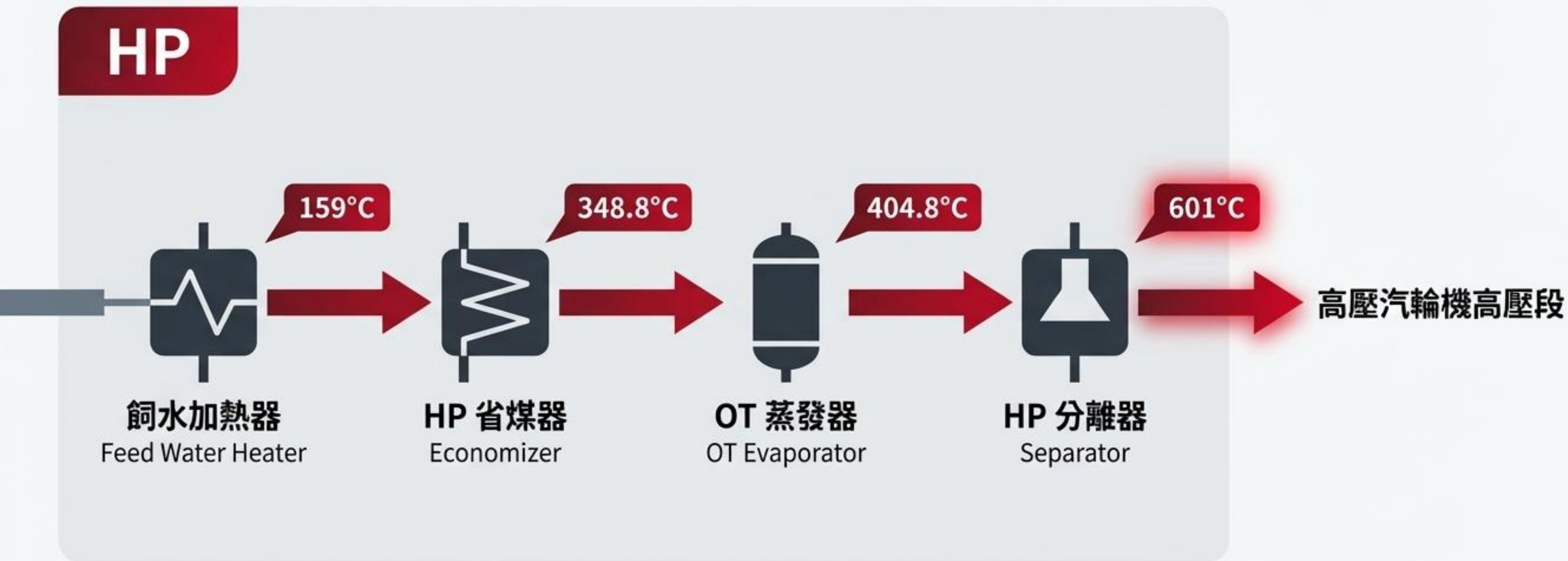
廢熱回收的本質是一場熱能的極致萃取。系統攔截高達 706°C 的工業廢氣，透過內部的熱交換機制將熱能轉換為機械動力 (Power)，最終將排放氣體溫度大幅降低至 74°C，實現能源的二次利用。

系統架構：階梯式的三階段熱能萃取 (Three-Stage Architecture)



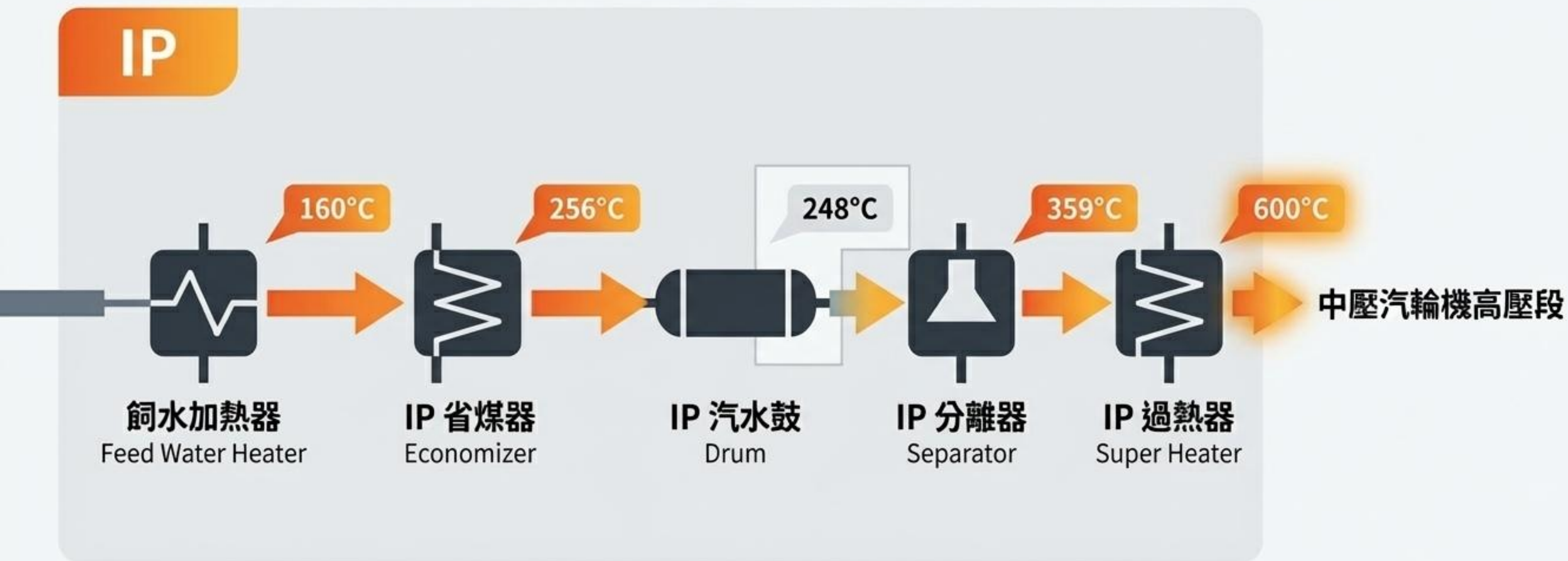
為了最大化回收效率，氣流依序穿過高壓 (High Pressure)、中壓 (Intermediate Pressure) 與低壓 (Low Pressure) 三個獨立的熱交換模組。透過精確的壓力分級，系統能在不同的溫度區間內，將每一分熱能的利用率推向極限。

高壓段 (HP) 流程：捕捉最高溫的熱能 (High Pressure Loop)



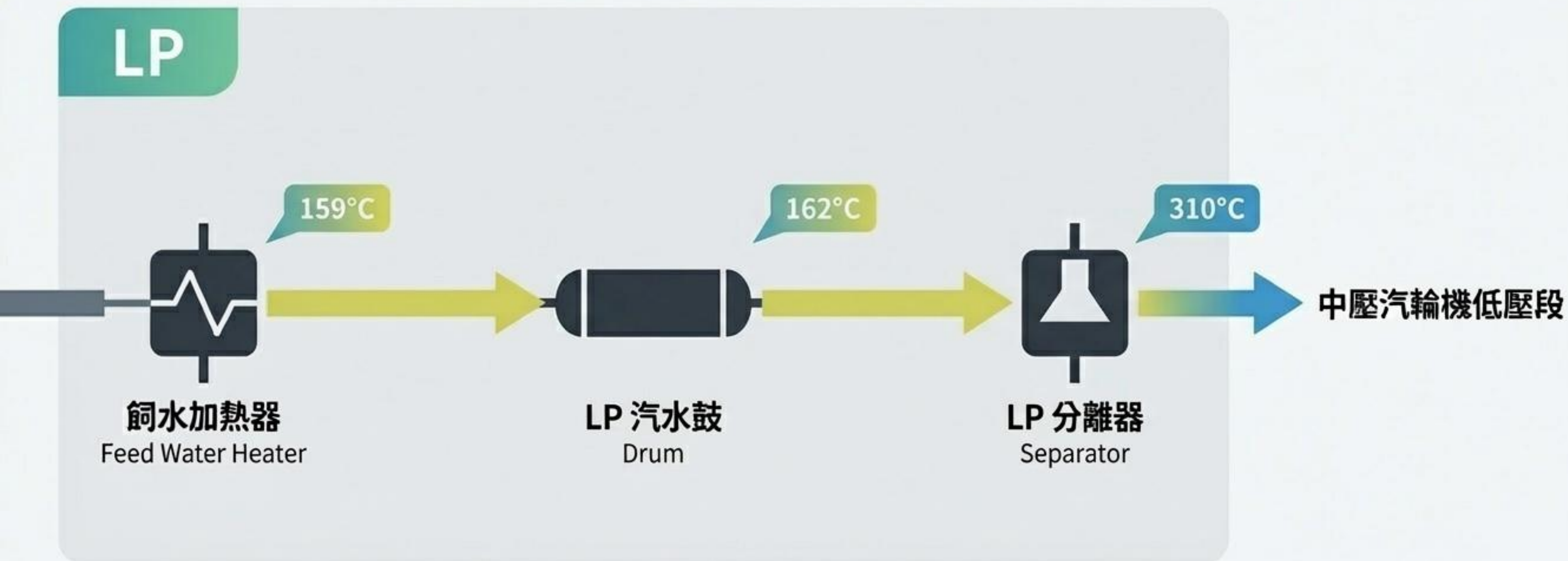
高壓迴路直接吸收氣流前端的最強熱能。經過四個階段的連續加熱與相變，最終穩定輸出高達 601°C 的高壓蒸汽，作為系統中最核心的動力來源。

中壓段 (IP) 流程：二次能量轉化與過熱機制 (Intermediate Pressure Loop)



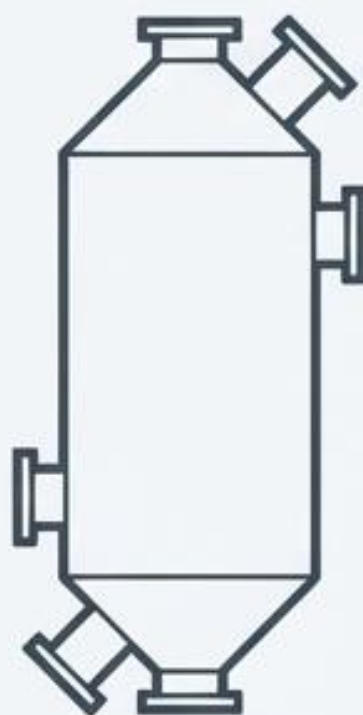
在中壓迴路中，流體進入汽水鼓（Drum）進行氣水分離。為了滿足汽輪機的需求，從分離器產出的 359°C 蒸汽必須再通過「IP 過熱器」，一口氣拉升至 600°C，完成二次高質量動能的轉化。

低壓段 (LP) 流程：榨乾尾端的剩餘熱值 (Low Pressure Loop)



作為熱能回收的最後一道防線，低壓迴路採用相對精簡的架構。透過汽水鼓與分離器，成功將 159°C 的飼水轉化為 310°C 的可用蒸汽，導入中壓汽輪機低壓段，徹底榨乾系統餘熱。

核心設備構型對比：OT 蒸發器 vs. 汽水鼓 (Equipment Anatomy)



OT 蒸發器 (OT Evaporator)

專用於 HP (高壓) 段



汽水鼓 (Drum)

用於 IP (中壓) 與 LP (低壓) 段

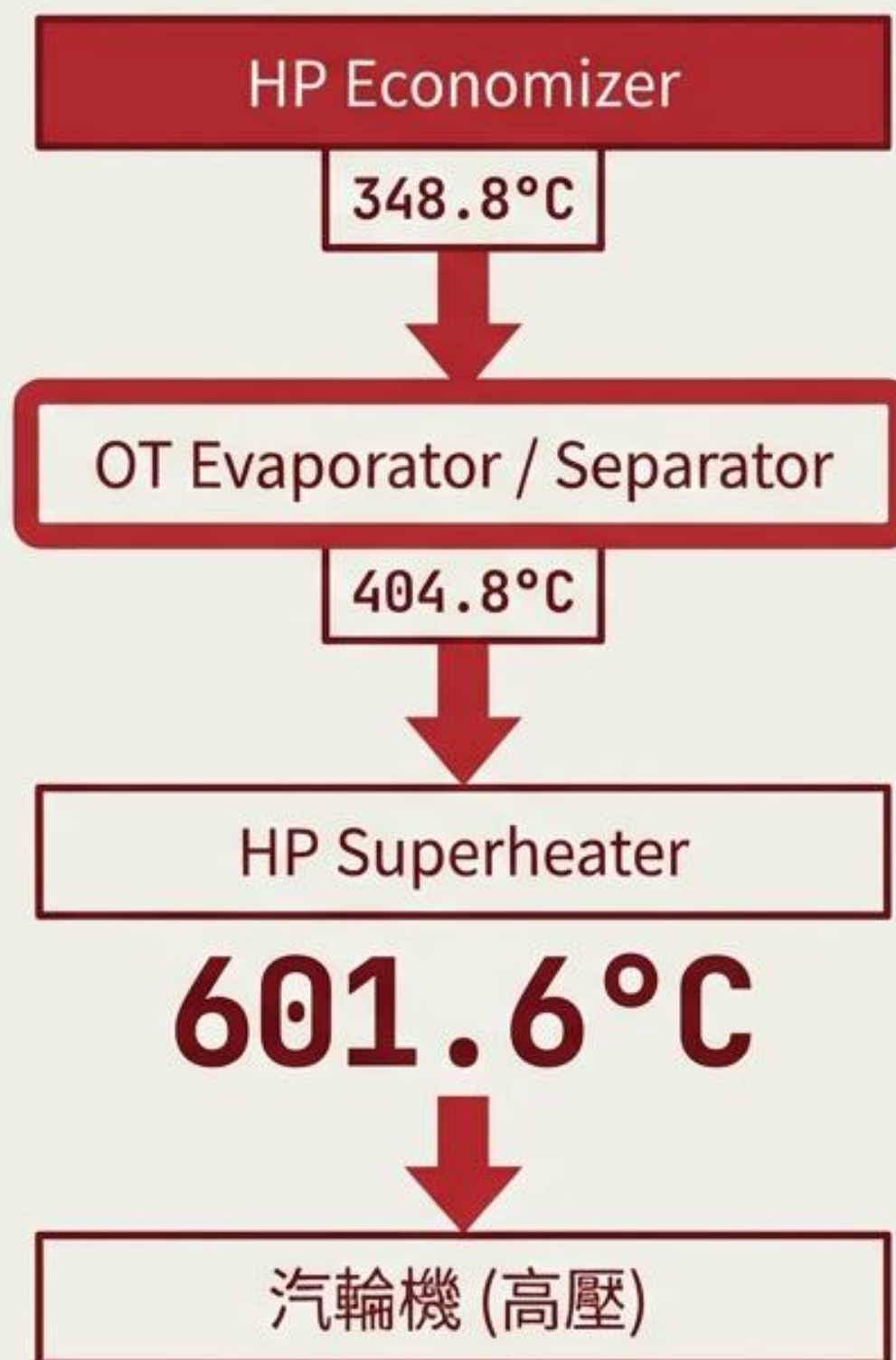
不同的壓力層級造就了不同的硬體構型。高壓段採用直立式的 OT 蒸發器以承受極端高壓並進行快速熱交換；而中、低壓段則採用橫式的汽水鼓，提供更大的內部空間來確保氣相與液相能徹底分離。

高壓系統 (HP System)

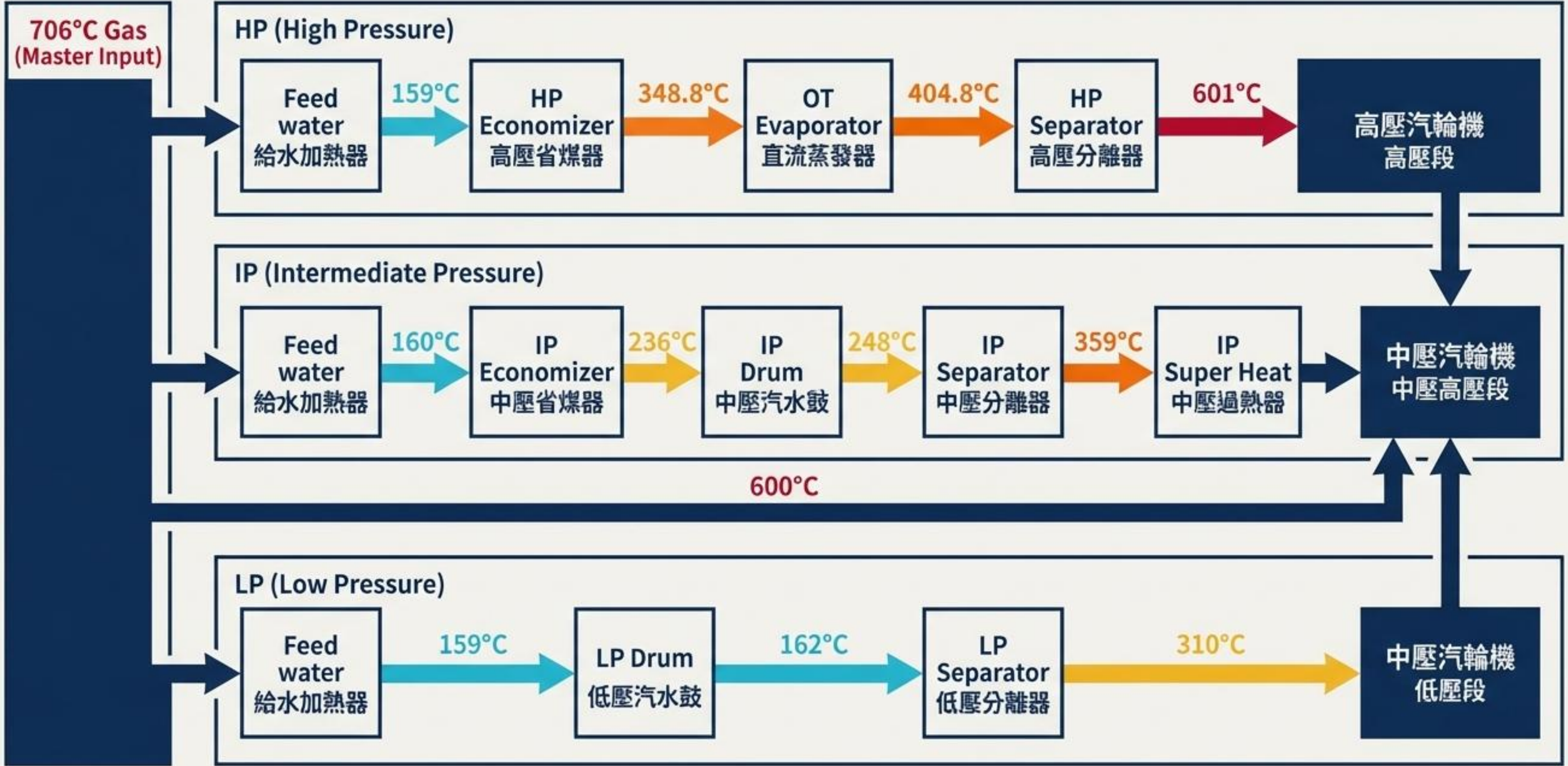
— 貫流式技術應用

最高溫循環 (Highest Temperature Cycle):
601.6°C

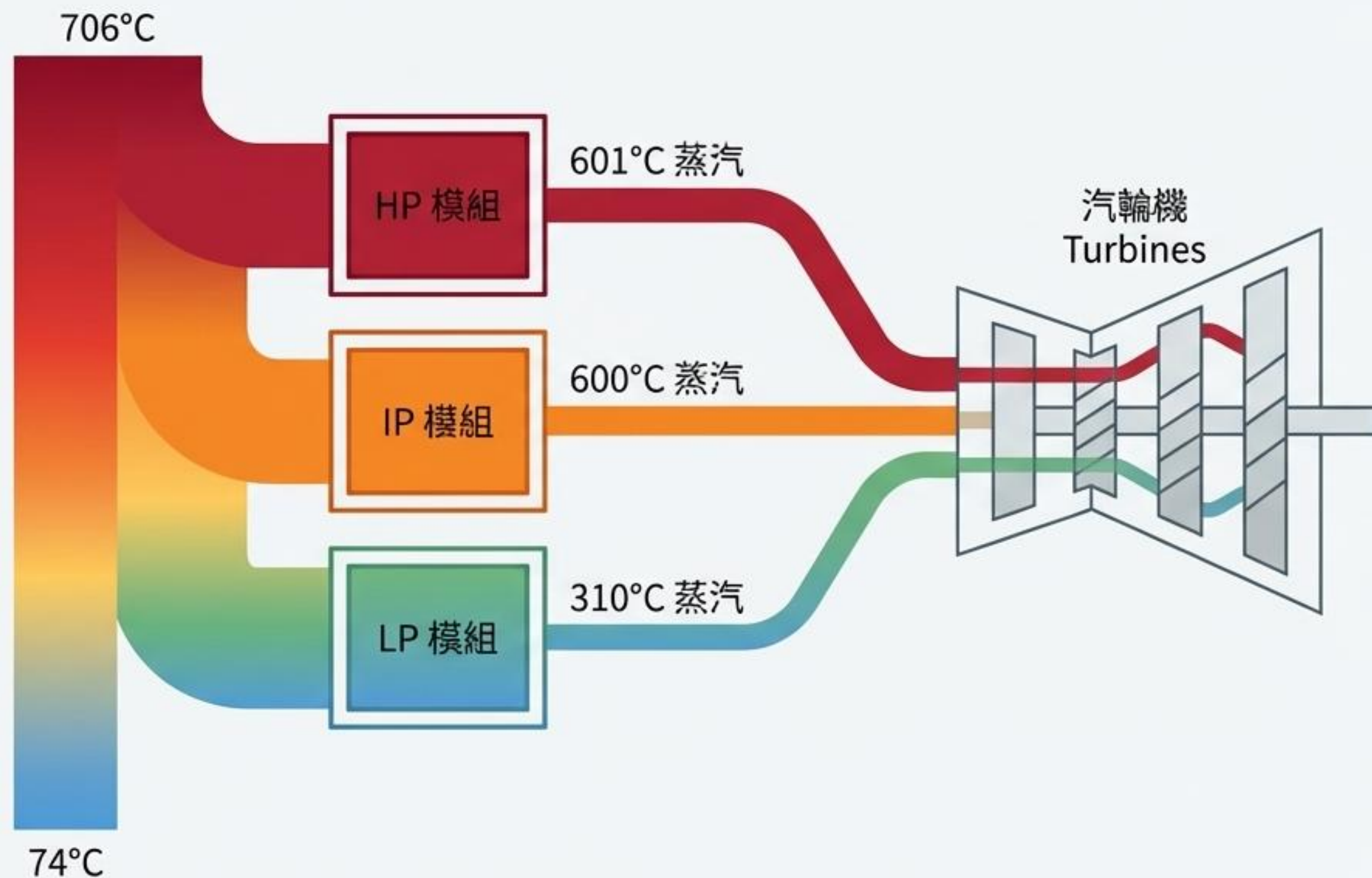
核心技術 (Core Tech):
OT (Once-Through) Evaporator / Separator
- 取代傳統汽鼓設計



系統全貌：三階段熱能回收矩陣 (The Complete Process Map)



系統總覽：廢熱回收的極致工藝 (The Complete Thermal Topography)



透過 HP、IP、LP 三條獨立卻又彼此協同的水汽循環路線，系統完美地將單一的極高高溫氣流，轉化為三個不同壓力層級的強大動力來源。這不僅是溫度的降低，更是能源利用效率的極致展現。

核心設備功能矩陣與總結 (Component Glossary & Synthesis)



省煤器 (Economizer)

捕捉中低溫餘熱，執行進入相變前的關鍵水預熱。



蒸發器 (Evaporator)

系統核心吸熱區，促使液態水吸收大量潛熱轉為蒸汽。



汽水鼓/分離器 (Drum/Separator)

精準分離氣液雙相，確保進入汽輪機的流體品質。



過熱器 (Super Heat)

進一步加熱乾蒸汽，極大化焓值與最終發電推動力。

透過嚴密的 HP、IP、LP 三段式熱力學設計，本熱回收系統
成功將每一分廢氣潛能，轉化為無可挑剔的發電效率。

流程邏輯回顧與註記

系統總結 (System Summary)

- ⚙️ 多壓循環 (Multi-Pressure) : **LP**, **IP**, **HP** 三種壓力級別同步運作。
- ⇒ 再熱優化 (Reheat Optimization) : **IP** 蒸汽導入再熱器，達成 **600.5°C** 高效能輸出。
- 🔗 貫流技術 (Once-Through) : **HP** 系統採用 OT 蒸發器，達成最高溫 **601.6°C**。

圖表註記 (Diagram Notes)

- 📋 所有溫度數據源自現場流程圖。
- 📖 ECO. = Economizer (省煤器)。



鍋爐及壓力容器安全規則

第十七條 雇主對於鍋爐之安全閥及其他附屬品，應依下列規定管理：

- 一、安全閥應調整於最高使用壓力以下吹洩。但設有二具以上安全閥者，其中至少一具應調整於最高使用壓力以下吹洩，其他安全閥可調整於超過最高使用壓力至最高使用壓力之一點零三倍以下吹洩；具有釋壓裝置之貫流鍋爐，其安全閥得調整於最高使用壓力之一點一六倍以下吹洩。經檢查後，應予固定設定壓力，不得變動。
- 二、過熱器使用之安全閥，應調整在鍋爐本體上之安全閥吹洩前吹洩。
- 三、釋放管有凍結之虞者，應有保溫設施。



GE Power

Non-Public

Spec. No. T219DA
Identification Number: 1GP033971 Rev A
Date: 5/06/2021

FEEDWATER PUMP DESIGN

Each pump must be designed to operate at all conditions defined below.
Assumptions made by GE related to pump and piping/valve pressure drop shall be verified by the customer.

1. Potential Pump Rating Points

		Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7	Case 8
Load case name in the Datablock		NEW_Max Throttle Pressure - HPFH	NEW_Full ST Bypass - Max HPFH Flow - common LP	NEW_Cold Dry - 20T - RR 1st Start - No T.A - common LP	NEW_Hot Dry - MECL - 10T - STOn	NEW_BC - 65% CC	Expected	Not in Datablock: 40% flow of Case 1	Not in Datablock: 100% flow of max pump suction flow case
Case name as per GE design guideline		ST Operation, Max operating pressure at HP ST Inlet	ST Bypass, Max suction massflow	RR Start-up, Max suction massflow	MECL, Min suction massflow	65%CC	Best Efficiency Point	SV Blowing	Max Case
Normal HP Discharge Flow per Pump	(kg/s)	136.1	134.9	149.7	73.7	200.1	133.9	58.9	149.7
Make-up Flow for Blowdown [Note 7]	3%	3%	3%	3%	0%	0%	0%	0%	3%
HP Pump Flow Margin****	5%	5%	5%	5%	0%	0%	0%	0%	45%
HP Discharge Massflow per Pump	(kg/s)	147.2	145.9	161.5	73.7	200.1	133.9	58.9	222.2
Suction Volume Flow, HP Flow Fraction per Pump	(m³/hr)	585.7*	576.6*	631.3*	285.5*	783.6*	530.4*	234.3*	666.7*
Normal IP Discharge Flow per Pump	(kg/s)	14.1	34.7	30.1	5.4	16.2	10.1	6.1	30.1
Make-up Flow for Blowdown [Note 7]	3%	3%	3%	3%	0%	0%	0%	0%	3%
IP Pump Flow Margin****	5%	5%	5%	5%	0%	0%	0%	0%	26%
IP Discharge Massflow per Pump	(kg/s)	15.2	38.9	32.0	5.4	16.2	10.1	6.1	38.5
Suction Volume Flow, IP Flow Fraction per Pump	(m³/hr)	60.6*	145.6*	125.3*	21.1*	63.6*	38.8*	24.3*	150.3*
Total Suction Volume Flow per Pump	(m³/hr)	646.4*	722.5*	756.5*	306.6*	847.2*	570.3*	258.5*	819.1*
HP OT CV Outlet Pressure	(bars)	209.2	205.8	198.0	100.1	148.6	204.9		
HP OT CV Outlet Pressure Variation Allowance*****	5.0%	5.0%	5.0%	5.0%	0.0%	0.0%	0.0%		
Maximum Operating Pressure HP OT CV Outlet	(bars)	219.7	216.1	207.9	100.1	148.6	204.9		
Maximum sustained operating pressure at boiler inlet (bars)								239.0	
Total Dynamic Head HP Discharge	(m)	2486*	2509*	2438*	802*	1441*	2246*	2537*	**
IP Drum Pressure	(bars)	40.4	40.8	38.6	17.7	29.2	38.7		
IP Drum Pressure Variation Allowance*****	5.0%	5.0%	5.0%	5.0%	0.0%	0.0%	0.0%		
Maximum Operating Pressure IP Drum	(bars)	42.4	42.8	40.8	17.7	29.2	38.7		
IP Drum Pressure when SVs Blowing	(bars)							58.7	
Minimum Total Dynamic Head Interstage Bleed	(m)	>=518*	>=588*	>=664*	>=129*	>=230*	>=381*	>=502*	**
Water Temperature at Suction	(°C)	163.8	156.9	146.6	139.3	149.5	159.7	163.6	146.6
Water Pressure at Suction Flange	(bars)	21.6	18.1	15.3	40.7	33.3	24.7	22.0	14.6
Specific Volume at Suction	(m³/kg)	0.00111	0.00110	0.00109	0.00108	0.00109	0.00110	0.00111	0.00109
Density at Suction	(kg/m³)	904.6	911.2	920.8	928.8	919.2	908.9	904.9	920.8
NPSH Available at Pump Foundation Level	(m)	167.6***	115.9***	120.6***	407.9***	317.7***	208.5***	174.8***	105.5***
NPSH _{req} foundation level [Note 4]									
NPSH Required	(m)	**	**	**	**	**	**	**	**
Pumps Required:									
Total Number Installed per Block		3	3	3	3	3	3	3	3
Total Number Operating per Block		2	2	2	2	1	2	2	2

Other requirements [Note 2]:

Pump HP discharge:

Max allowable pump shut-off pressure (at zero flow) (bars) 341
Max allowable pump shut-off head (m) 3483

Pump interstage bleed:

Max allowable pump shut-off pressure (at zero flow) (bars) 87
Max allowable pump shut-off head (m) 890

Pump suction:

Max pressure at suction flange [Note 6] (bars) 60.0

* Pump Design Requirement

** To be provided by pump supplier

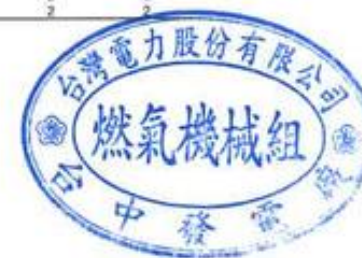
*** GE estimate, assuming pump foundation level at floor level. Final values to be provided by EPC based on suction piping arrangement

**** Wrapped design margin on pump flow to cover pump aging and higher steam production due to GT manufacturing tolerances

***** Wrapped allowance on pump head to cover mainly:
- ST manufacturing tolerances resulting to higher steam pressure at ST inlet than shown on HBO & Datablock
- GT manufacturing tolerances resulting to higher GT exhaust energy than shown on HBO & Datablock
- Pump manufacturing tolerances on rated head

Notes:

- The pump head curve must be continuously rising with a shutoff head at least 115%, but not more than 135% of rated head.
- Maximum head rise/discharge pressure limitations are based on BOPNRSQ design pressure limits.
- Pump best efficiency point shall be close to the best efficiency point while still respective of the head rise limitations.
- The ratio of the available NPSH and the required NPSH must fulfill the following criteria at pump center line level
 $NPSH_{av} \geq 1.4 \times NPSH_{req}$
- Minimum continuous flow shall preferably not be higher than 40% of Case 1 flow.
- For pump control and protection design purposes, vendor shall provide the pump characteristic for full operating range. Speed will be limited to not exceed the system design pressure.
- Make-up massflow is based on a percentage from Case 1.
- Assuming max LP ECO design pressure.





GE Power

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Spec. No. T219DA
Identification Number: 1GP033971 Rev A
Date: 5/06/2021

2. Pump Design Basis Data

Discharge pressure is based on the following assumptions:

The Design Point HP Discharge Pressure is based on the following assumptions:

	Case 1	Case 2	Case 3
HP discharge pressure to HP OT Separator			
Maximum Sustained (Normal) Operating HP OT Separator Pressure	2474.8 (m)	2417.1 (m)	2301.5 (m)
HP Economizer Pressure Drop (incl. HRSG piping)	113.8 (m)	129.1 (m)	154.3 (m)
Total HP FW Discharge Piping Pressure Drop up to HRSG TP (including Automatic Recirculation Valve)	45.4*** (m)	44.8*** (m)	48.9*** (m)
Feedwater Control Valve Pressure Drop	28.2 (m)	28.0 (m)	27.7 (m)
Elevation Head (from top of pump foundation to HP OT Separator bottom)	38.0 (m)	38.0 (m)	38.0 (m)
Suction Line Pressure Drop (pipe incl. filter)	10.6*** (m)	12.7*** (m)	14.1*** (m)
Suction Pressure (at FW suction flange)	243.7 (m)	179.8 (m)	168.9 (m)
Margin on dynamic losses and static head	15%	15%	15%
Total Dynamic Head	2485.8 (m)	2508.6 (m)	2437.8 (m)

The Design Point IP Discharge Pressure is based on the following assumptions:

	Case 2	Case 3
IP discharge pressure to IP drum		
Maximum Sustained (Normal) Operating IP Drum Pressure	479.4 (m)	449.0 (m)
IP Economizer Pressure Drop (incl. HRSG piping)	84.2 (m)	174.9 (m)
Total IP FW Discharge Piping Pressure Drop up to HRSG TP	17.0*** (m)	12.5*** (m)
Feedwater Control Valve Pressure Drop	33.6 (m)	33.2 (m)
Elevation Head (from top of pump foundation to IP drum bottom)	38.0 (m)	38.0 (m)
Suction Line Pressure Drop (pipe incl. filter)	12.7*** (m)	14.1*** (m)
Suction Pressure (at FW suction flange)	179.8 (m)	168.9 (m)
Margin on dynamic losses and static head	15%	15%
Total Dynamic Head	493.3 (m)	522.5 (m)

or:

	Case 2	Case 3
IP discharge pressure to GT fuel gas efficiency preheater		
Maximum Fuel Gas Supply Pressure at FGPH Inlet	528.0 (m)	522.5 (m)
Minimum Pressure Difference Between Water and Fuel	22.4 (m)	22.1 (m)
FGPH Water Side Pressure Drop	44.2 (m)	42.7 (m)
IP FW Piping Pressure Drop from HRSG IP ECO Outlet up to FGP Inlet	35.5*** (m)	12.4*** (m)
IP Economizer Pressure Drop (incl. HRSG piping)	84.2 (m)	174.9 (m)
Total IP FW Discharge Piping Pressure Drop up to HRSG TP	17.0*** (m)	12.5*** (m)
Elevation Head (from top of pump foundation to FGPH)	8.0*** (m)	8.0*** (m)
Margin on dynamic losses and static head	15%	15%
Total Dynamic Head	587.9 (m)	663.9 (m)

or:

	Case 2	Case 3
IP discharge pressure to HP bypass water injection valve		
HRH Steam Pressure Downstream HP Bypass Spray	455.0 (m)	432.0 (m)
HP Bypass Spray CV (incl. spray nozzle)	134.2 (m)	132.8 (m)
IP FW Piping Pressure Drop from HRSG TP up to HP Bypass TCV Inlet	44.8*** (m)	10.0*** (m)
Total IP FW Discharge Piping Pressure Drop up to HRSG TP	17.0*** (m)	12.5*** (m)
Elevation Head (from top of pump foundation to HP bypass spray)	20.0*** (m)	20.0*** (m)
Margin on dynamic losses and static head	15%	15%
Total Dynamic Head	503.4 (m)	444.8 (m)

or:

	Case 3
IP discharge pressure to HRSG HRH steam interstage desuperheater water injection valve	
HRH Steam Pressure Downstream Interstage Spray	408.5 (m)
Atomizer Spray Nozzle, CV & Piping Pressure Drop	143.9 (m)
Total IP FW Discharge Piping Pressure Drop up to HRSG TP	12.5*** (m)
Elevation Head (from top of pump foundation to HRH interstage desuperheater)	38.0 (m)
Margin on dynamic losses and static head	15%
Total Dynamic Head	441.6 (m)

*** GE estimate, final values to be provided based on EPC arrangement





中華民國110年5月18日

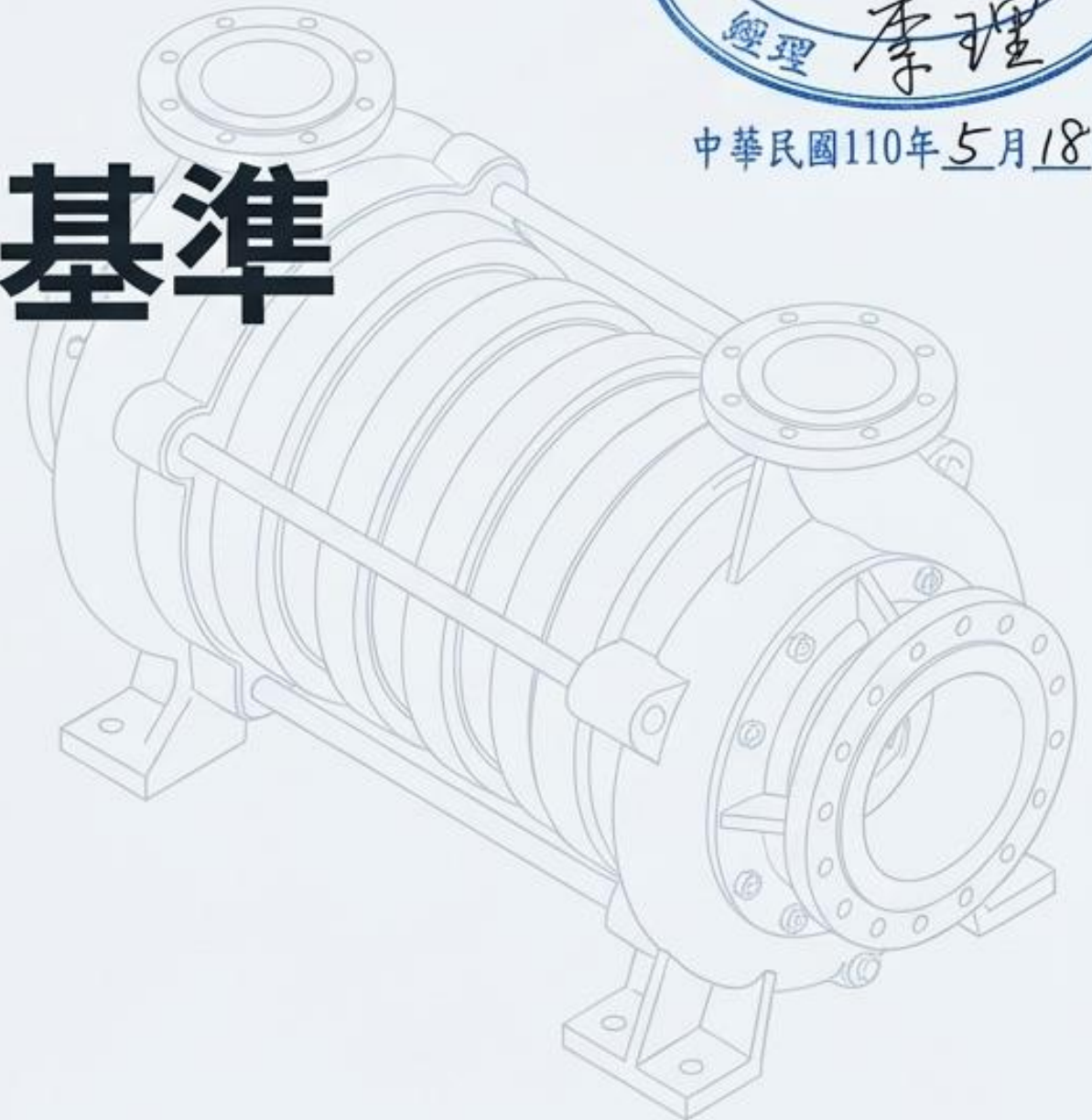
GE 飼水泵系統設計基準 與運轉包絡線解析

系統參數、總動態水頭 (TDH) 計算與
運轉極限條件 (Spec. No. T219DA)

文件編號 (Identification Number): 1GP033971 Rev A

日期 (Date): 5/06/2021

設備 (Equipment): 飼水泵 (Feedwater Pump)



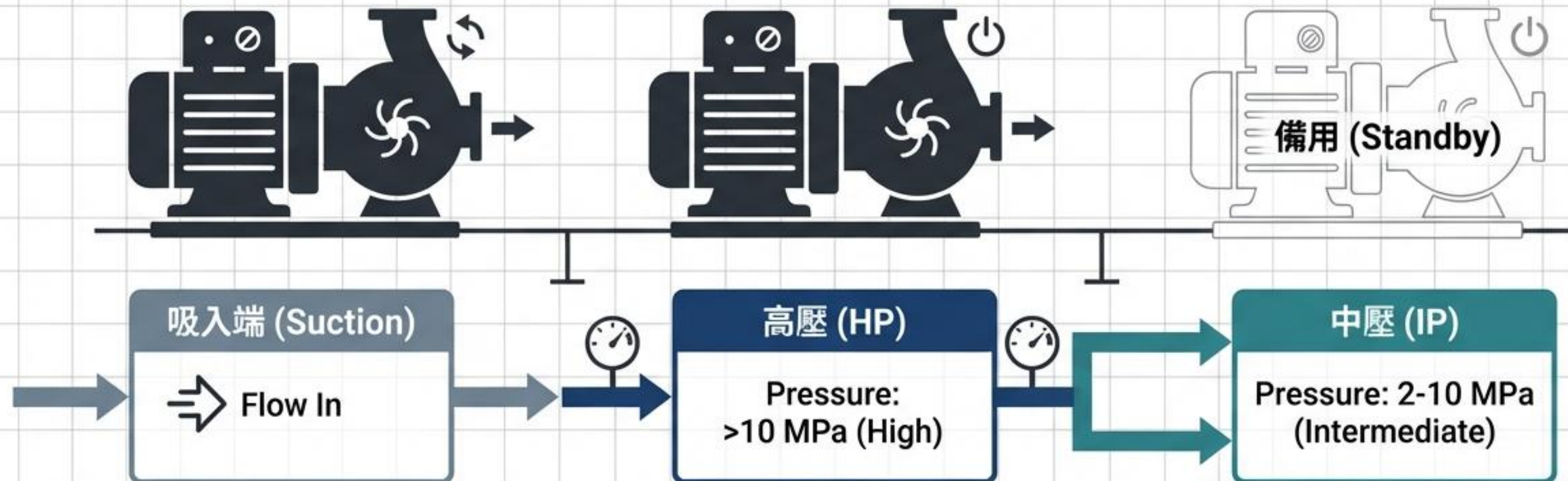
系統配置與硬體基準範疇

KPI

機組配置 (Pumps Required) :
每部機組安裝 3 台

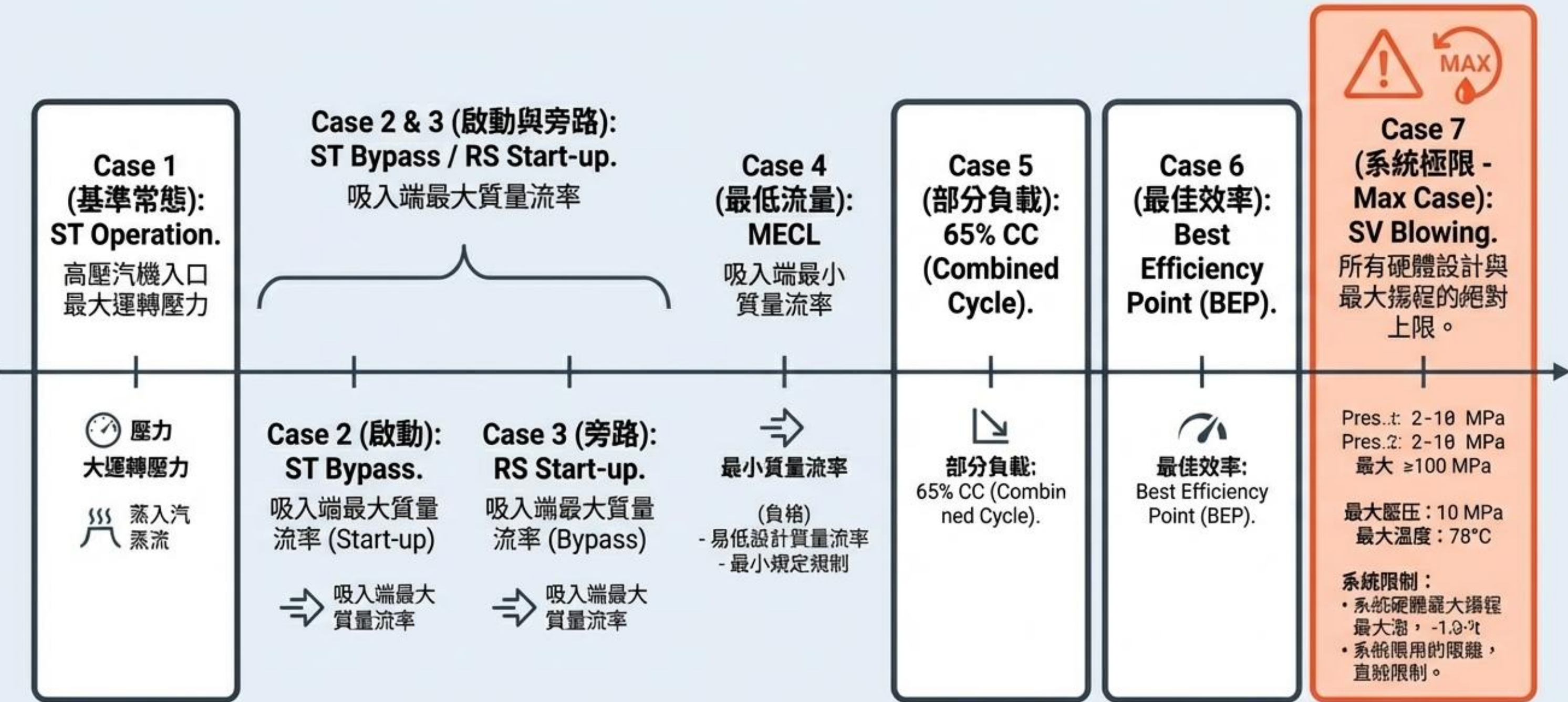
KPI

運轉模式 (Operation) :
每部機組同時運轉 2 台



設備定位 (Role) : 負責將水流從吸入端 (Suction) 加壓並分配至高壓 (HP) 與中壓 (IP) 系統，涵蓋汽包 (Drum)、預熱器與減溫器等關鍵節點。

七大運轉情境定義：從常態到極限 (Cases 1-7)



基準資料來源：系統設計規範矩陣

總動態水頭 (TDH) 設計基準解構



說明 (Description): 飼水泵的排放壓力設計並非單一數值，而是根據不同目的地背壓 (Backpressure)、管路摩擦損失以及高程變化進行疊加計算。



三大核心排放路徑 (Three Core Discharge Paths):

1. HP OT Separator (高壓端)
2. IP Drum (中壓汽包)
3. IP 特殊注入點 (預熱器、旁路注水閥、減溫器)

共通原則 (Universal Rule): 所有動態損失與靜態水頭計算均強制加上 15% 餘裕 (Margin on dynamic losses and static head) 以確保系統安全性。

NPSH 限制與流體物理特性

$$\text{NPSHa} \geq 2.1 \times \text{NPSHr}_{3\%}$$

核心安全規範：可用淨正吸入水頭 (NPSHa) 必須至少為泵浦所需淨正吸入水頭 (NPSHr) 的 2.1 倍，確保系統不發生空穴現象 (Cavitation)。

可用水頭變化 / NPSHa Variation

Case 1 (常態)：167.8 m

Case 7 (極限)：降至 108.5 m

流體密度 / Density at Suction

Case 1 (常態)：911.2 kg/m³

Case 7 (極限)：926.8 kg/m³

說明：密度隨情境波動，直接影響泵浦做功效率。

設備安全極限與性能保證條件



最大容許極限 (Maximum Allowable Limits)

- 零流量時最大關閉壓力 (Max allowable shut-off pressure at zero flow): **341 barg / 3463 m.**
- 吸入法蘭最大壓力 (Max pressure at suction flange): **60.0 barg.**



連續運轉限制 (Continuous Operation)

- 最小連續流量不得高於 Case 1 流量的 **40%** (Minimum continuous flow shall not be higher than **40%** of **Case 1** flow).



效率保證 (Efficiency Guarantee)

- 最佳效率點 (BEP) 必須貼近最頻繁的運轉點 (Case 6)。
- 揚程曲線 (Head curve) 必須連續上升，關閉水頭至少為額定揚程的 115%，但不得超過 135%。

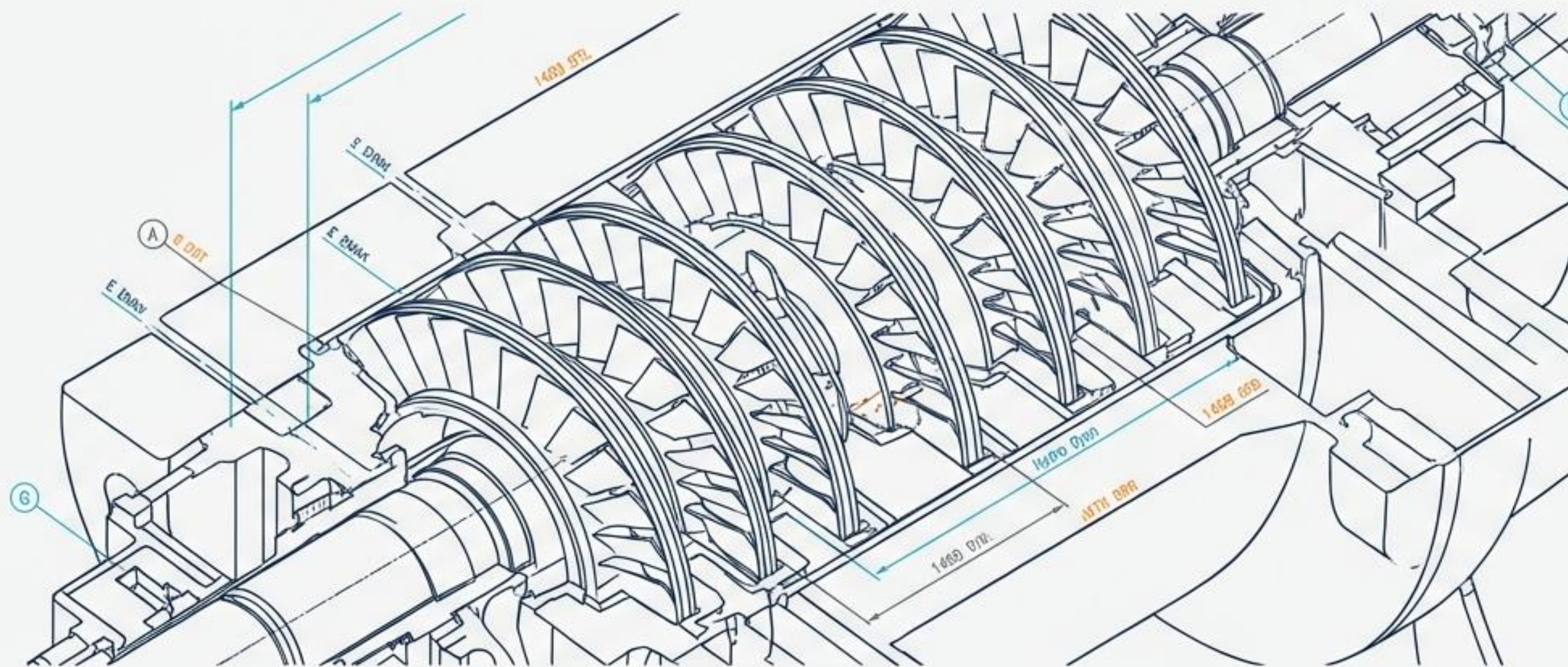


設計邊界 (Design Envelope)

- 補水質量流率 (Make-up massflow) 以 Case 1 的百分比為基準。
- 供應商需提供完整轉速範圍之泵浦特性曲線供系統設計檢核。

多級式可變速中高壓聯合給水泵技術分析

台灣電力公司 Pool 3 專案：E6+5 型給水泵設計詳解

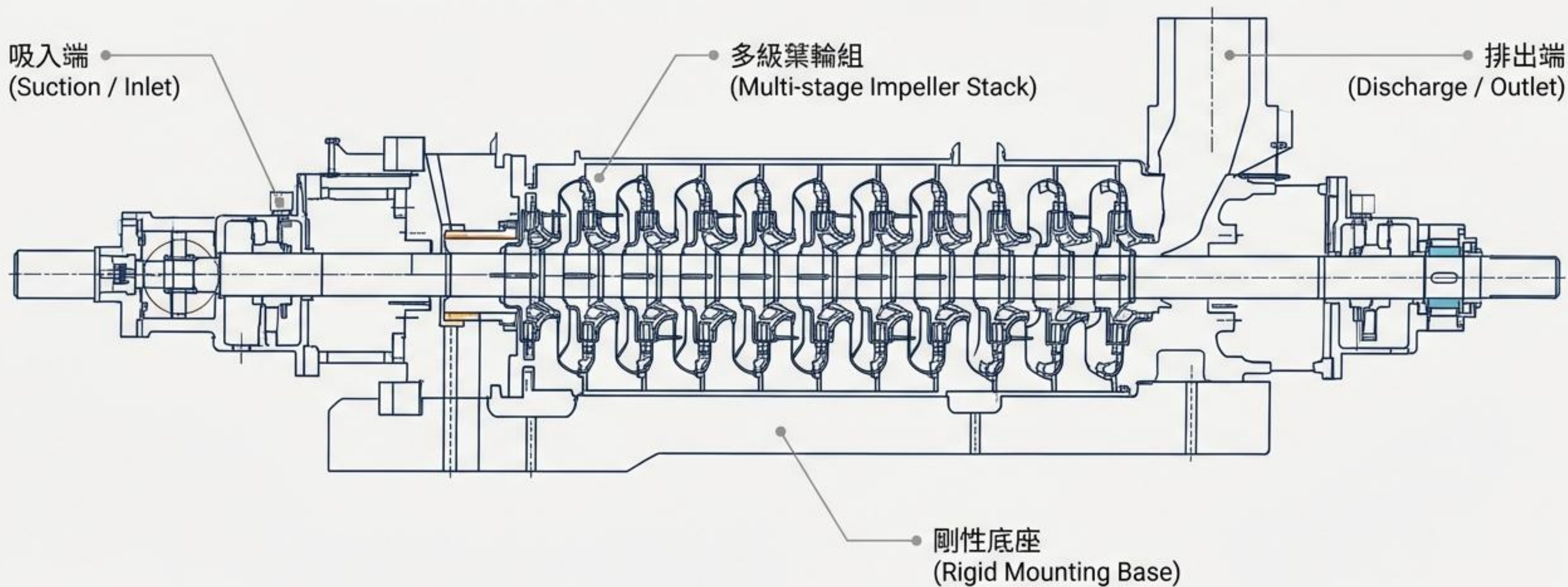


APPLICATION: HP / IP BOILER FEED WATER (高壓/中壓鍋爐給水)

PROJECT REF: TAIWAN POWER COMPANY POOL 3

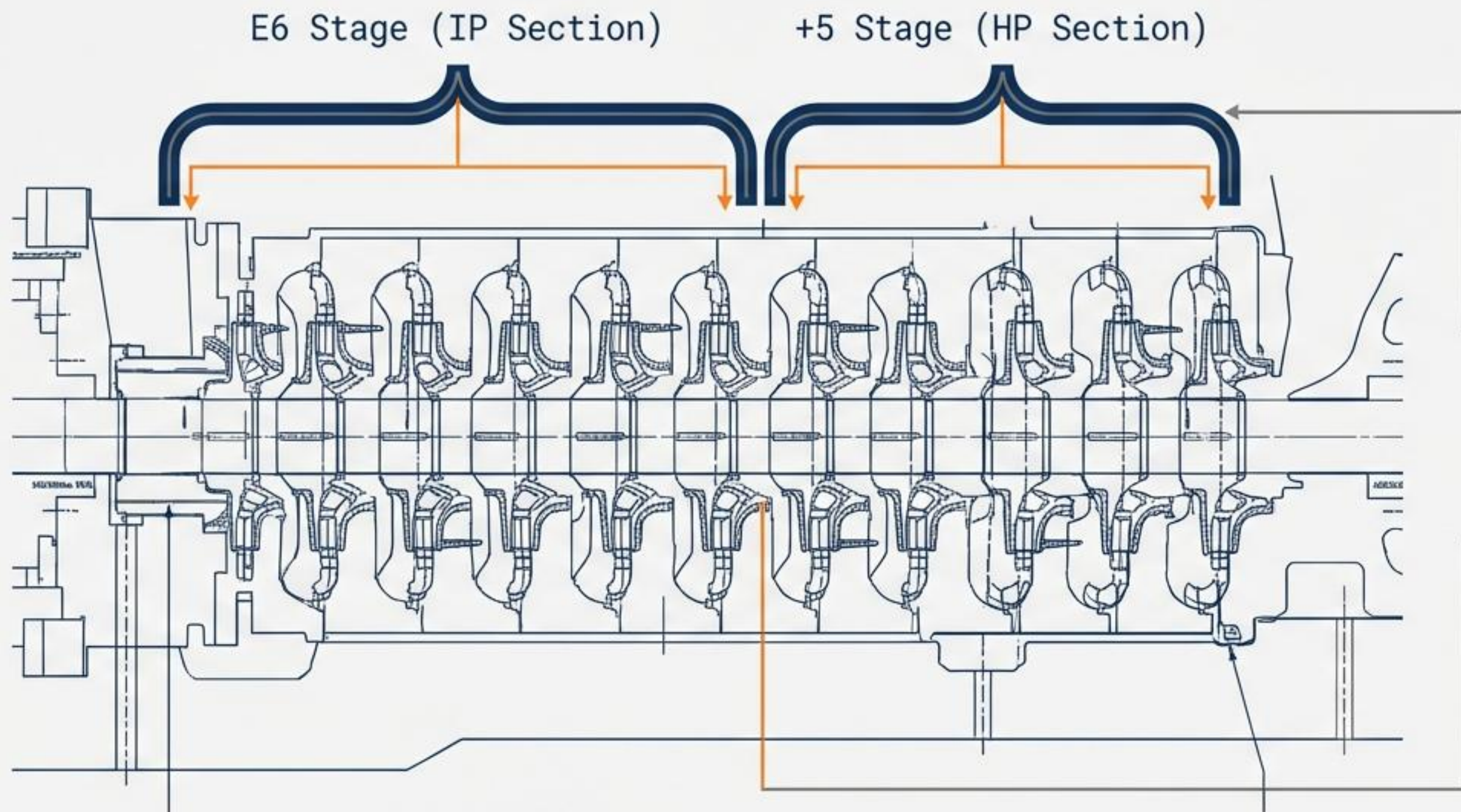
Noto Sans TC

系統總覽：縱剖面構造



Noto Sans TC: The assembly demonstrates a robust, elongated shaft design necessary to accommodate 11 compression stages within a single housing.

核心機制：E6+5 多級壓縮配置



11 級增壓

Fluid passes through 11 distinct impellers.

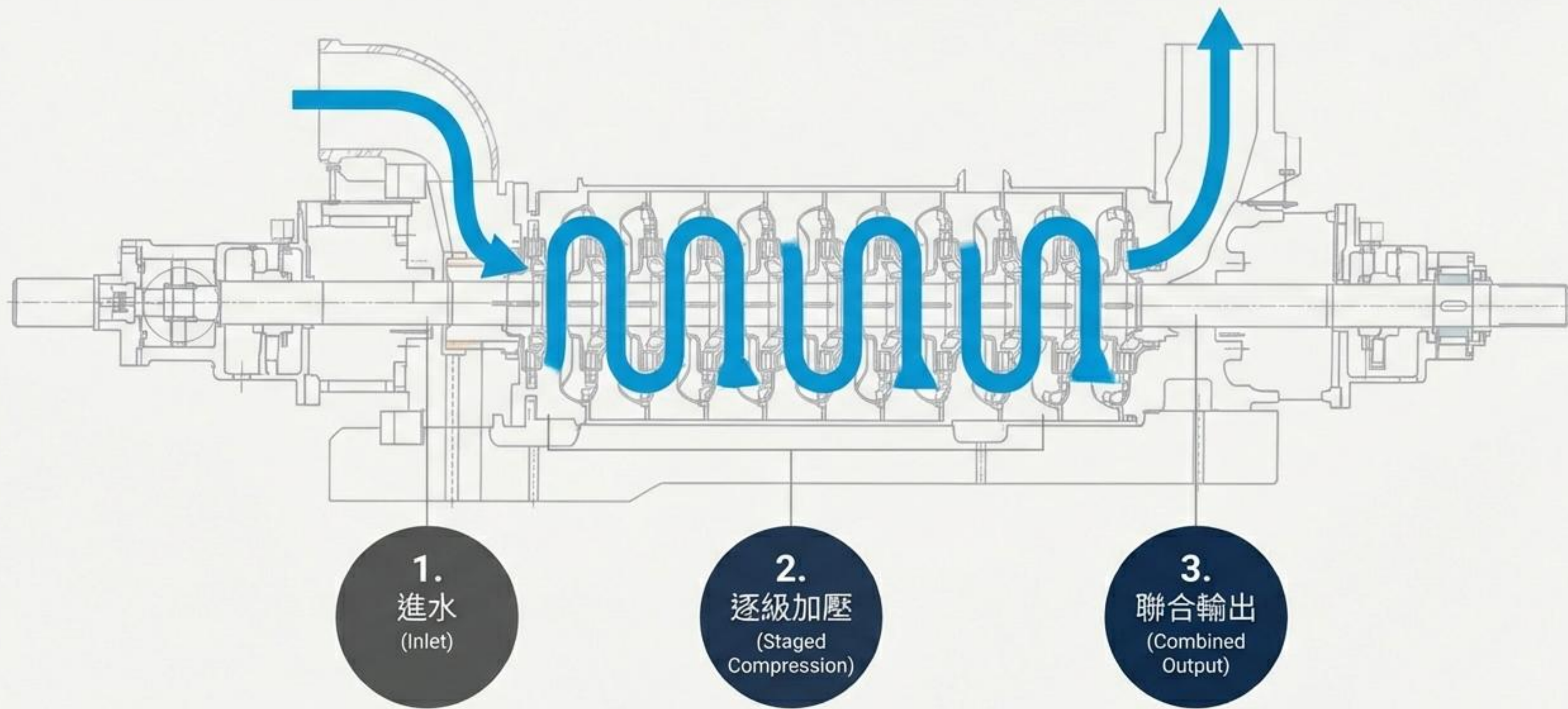
分組配置

6 Stages + 5 Stages combined configuration.

流體導葉

Stationary diffusers convert velocity to pressure.

流體動力學路徑





技術總結 (Technical Summary)

高性能 (High Performance)

11-Stage (E6+5) configuration for optimal head.

高可靠性 (High Reliability)

Thrust Tilt Pad Bearings ensure axial stability.

高耐用度 (High Durability)

Chrome Steel internals for long service life.

NOTE: VALUES OF CLEARANCES FOR 4-VALVE GRL MAIN FORCED ON INSTRUCTION MANUAL

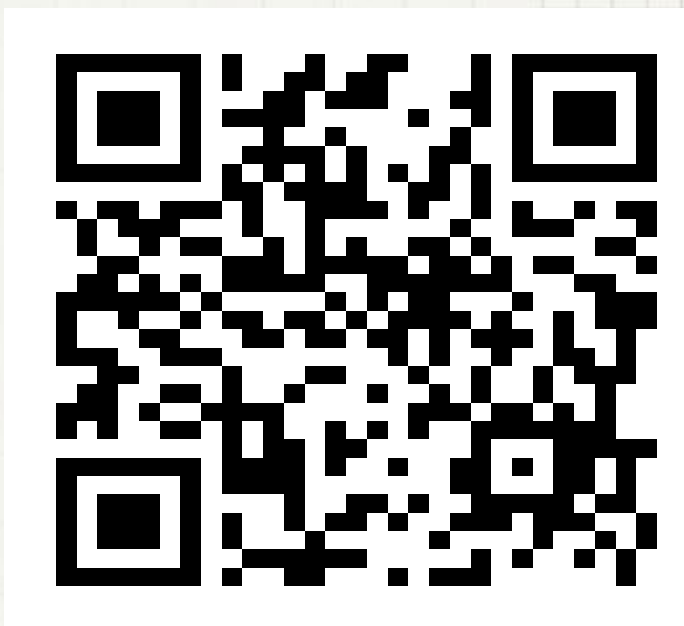
Design

Specifications.



謝謝 (Thank You)

Q & A



Roboto Mono
Confidential Engineering Analysis